

LSTM-based QoS prediction for 5G-enabled Connected and Automated Mobility applications

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Abstract—Determining whether the network can provide the required resources -and as a result the required Quality of Service (QoS)- for Connected and Automated Mobility (CAM) applications is crucial, as human safety is involved. In order to identify potential QoS deterioration, Machine Learning, which is currently considered a momentous enabler for the 5G and beyond systems, can be exploited in order to timely predict such changes in the QoS and proactively notify the CAM application in order for it to flexibly adapt. This paper explores the validity and viability of Recurrent Neural Networks (RNNs) -and more specifically LSTMs- as potential enablers for realizing such predictions. To this end, a novel QoS prediction mechanism is presented that uses an LSTM architecture and network-related QoS metrics, in order to identify specific patterns and predict in advance the QoS that will be available in the near future to a Connected and Automated Vehicle (CAV). Simulation-based evaluation of the proposed scheme prove its potential gains for the CAM domain.

Index Terms—5G, V2X, CAM, Quality of Service, Prediction, Machine Learning

I. INTRODUCTION

Vehicle-to-Everything (V2X) communications are considered as one of the structural enablers for future Intelligent Transportation Systems (ITS) and Connected and Automated Mobility (CAM), in the context of which, a plethora of novel applications and use cases has been described. A number of enhancements for the 5G System (5GS) - both the 5G Core (5GC), as well as the 5G Radio Access Network (RAN)- have been standardized in recent releases, in order to address V2X communications-related requirements and challenges [1] [2] [3]. V2X and CAM use cases have been acting as key drivers for the 5GS, as they introduce ultra strict requirements in relation to different Quality of Service (QoS) metrics, with the most often addressed ones being the end-to-end delay, reliability as well as throughput Key Performance Indicators (KPIs). Several use cases for V2X scenarios have been introduced, both by 3GPP, namely Vehicles Platooning, Advanced Driving, Extended Sensors, Remote Driving and Vehicle QoS Support, as well as by the 5G Automotive Association [4], namely Tele-operated Driving (ToD), High-density Platooning, Hazardous Location Warning, Lane Merge, Software update and Infotainment.

Multiple Access/Mobile Edge Computing (MEC) will further facilitate the efficient performance of V2X and CAM

applications, via minimizing the distance between the system's computing capabilities and the vehicles. ETSI has already released a specification [5], which describes V2X requirements and key issues for MEC-enabled architectures, for four identified V2X use case groups, such as Safety, Convenience, Advanced Driving Assistance and Vulnerable Road User; additionally, a number of key issues are identified, such as mobility and Quality of Experience (QoE) support, low latency communication support with multi-operator operations and communication traffic coordination with vehicles.

At the same time, a major breakthrough is taking place in relation to the intelligentization of the next generation of the mobile and wireless system, integrating numerous Machine Learning (ML)- and Artificial Intelligence (AI)-driven components, responsible for diverse network management aspects, such as resource allocation, mobility management, security, etc. Besides the numerous solutions that have been developed during the last years [6] [7], 3GPP has also introduced a novel network component, namely the Network Data Analytics Function (NWDAF) [8], which will be the main entity that will be integrating ML and AI logic into the system and will be able to interact with other 5GC and 5G RAN network entities for different purposes, such as data collection based on subscription to events, slice-specific resource allocation, retrieval of information from data repositories, on demand provision of analytics to consumers, etc. For CAM and V2X communications, network reliability and QoS are crucial in order to address different requirements in terms of ultra low end-to-end latency, minimum uplink/downlink throughput, acceptable packets loss, etc. As a result, the gains from such an evolution towards an intelligent system is considered compelling.

Towards this direction, the 5G Automotive Association (5GAA) describes the concept of predictive QoS [9], which refers to the mechanisms enabling mobile networks to proactively provide notifications about predicted QoS changes to interested consumers. According to this white paper, although 5GS already provides mechanisms to fulfil the majority of the performance requirements of V2X and CAM applications, due to varying network conditions, there could still be circumstances, in which the 5GS is unable to deliver the QoS required, leading the application to experience an unantic-

pated QoS degradation. QoS degradation in turn will require certain application-specific adaptations, such as slowing down the vehicle, lowering the quality of vehicle-oriented video streams for tele-operation, bringing the vehicle to a location where it can be stopped in a controlled way, etc.

Predicting the QoS in a timely manner, exploiting advanced ML and AI techniques, towards proactive V2X application adaptation is thus the focus of this paper. The use case, on which the proposed mechanism focuses is the ToD. The latter sets strict requirements in terms of the uplink throughput, as well as the downlink delay, for supporting the vehicle's uplink video stream transmission and ensuring timely delivery of tele-operation commands from a tele-operation center towards the semi-autonomous vehicle, respectively. We introduce a QoS prediction scheme that is capable of predicting in advance such QoS metrics, building upon a Long Short-Term Memory (LSTM) architecture.

The paper is organized as follows: Section II presents related work from the literature. Section III describes the proposed mechanism, along with the network data features, which are processed. Section IV provides the results of the proposed scheme's evaluation, while Section V concludes the document.

II. RELATED WORK

Although the concept of QoS prediction -from a broader network perspective- is not new, QoS prediction in 5G networks, for V2X and CAM applications is an unexplored domain, with very limited existing works.

The work in [10] targets in predicting threshold-driven QoS classes by coping with the QoS prediction as a classification problem. A MEC-enabled architecture is assumed, while the QoS prediction is evaluated via two V2X applications, namely, Advanced Driving and Emergency Brake Light. A Neural Network (NN)-based approach is proposed, combined with a Maximum Dependency (MD) algorithm for feature selection. The authors focus on predicting the expected end-to-end delay and the results are compared to other ML solutions, namely a Recurrent Neural Network with Long Short-Term Memory neurons, a Random Forest, and a Support Vector Machine. According to the authors, no significant prediction accuracy gains are observed in the alternative solutions, able to justify the noticeable increased cost of training, compared to the much simpler NN.

In [11], the proposed scheme collects different radio-related measurements, such as the reference-signal-received-quality/power (RSRQ/RSRP), the signal-to-noise ratio (SINR), the channel quality indicator (CQI), the user past average throughput, delay, etc. and use them as inputs for QoS prediction, via an auto-regressive integrated moving average (ARIMA) model. The proposed scheme considers a typical urban scenario for C-V2X communication, namely Manhattan Grid. Although in most scenarios the ARIMA model performs relatively satisfactory, for high number of UEs, the accuracy of the model drops considerably.

Perhaps, the most related work available in literature, a latency-prediction framework is described in [12] that focuses

on QoS prediction for delay-sensitive V2X applications. The proposed scheme combines an LSTM network with a k-medoids clustering algorithm to predict the trackable portion of the data; additionally it exploits a combination of Epanechnikov Kernel and moving average functions for predicting data that behave like random noise. The evaluation shows that the specific approach reduces the prediction error to half of a standard deviation of the raw data.

Regarding other works, in [13], a Cognitive Neural Network Delay Predictor (CNNDP) for high speed mobility in 5G C-RAN cellular networks is proposed for predicting the time variations of the communication channel. The results demonstrate a significant enhancement in the data rate of the network with the proposed approach. The work in [14] studied the problem of application rate allocation over different radio interfaces, and addressed the issue of different delay requirements per application. The authors propose two online predictive algorithms in order to handle the intermittence of secondary interface(s). The proposed algorithms' performance is presented consistently near-optimal, when using small prediction windows. Last but not least, the work in [15] performs prediction-based resource allocation focusing on application data rate; the specific work proposes the Threshold Percentage Dependent Interference Graph (TPDIG) using a Deep Learning-based resource allocation algorithm for city buses mounted with moving small cells (mSCs). A comparative analysis of resource allocation approaches is presented, in terms of Resource Block (RB) usage and average achievable data rate of a mobile Internet of Things (mIoT)-mSC network; the performance evaluation results confirm the gains achieved by the proposed contribution.

As it can be inferred, so far there is minimal effort with regard to Deep Learning-based QoS prediction for V2X communications and CAM applications. The work in [10] illustrates a considerable difference than the proposed work in this paper, as it addresses the problem as a classification problem; in our case, we model the QoS prediction as a time series problem, in order to retain a more fine-grained approach. In [11], the authors employ a simple moving average regression approach (ARIMA), which -as claimed by the authors in the evaluation part of the scheme- fails to cope with high number of vehicles and more complex scenarios. The only work that applies an LSTM architecture for predicting QoS and is considered the most related to the proposed one is [12]. Nevertheless, we are interested in evaluating how a finely-configured LSTM architecture -as a standalone ML algorithm- is able to perform satisfactory results towards predicting key QoS metrics, rather than a more complex framework that combines numerous ML and statistical approaches. This is mainly due to the fact that both MEC-enabled, as well as decentralized architectures in general, call for simple and efficient algorithms, capable of producing high performance with the lower computing resources utilization possible.

III. QoS PREDICTION SCHEME

This section presents an LSTM-based Multivariate Multi-step Autoencoder scheme [16] for real-time QoS prediction for a particular prediction time horizon. The term *multivariate*, results from the fact that the proposed LSTM-based scheme considers a set of different ToD service and network-related features, which are listed below:

- location of the ToD vehicle (pos_x, pos_y)
- speed of the ToD vehicle in m/s
- distance of the ToD vehicle from cell in m
- cell load percentage in Resource Blocks (RBs)
- number of vehicles connected in the cell, in which the ToD vehicle is attached to
- uplink throughput (ul_thr) of the ToD vehicle at time t

The core functionality of the proposed scheme is to predict the uplink throughput of a ToD vehicle on multiple time-steps into the future (*multi-step*), using multiple prior time-steps of each input feature (historical information). In this work, a time series model is used, since all the aggregated data points are equally distributed and indexed in time. The selected LSTM Autoencoder is an implementation of an Encoder-Decoder LSTM architecture and is designed to efficiently perform a sequence-to-sequence (seq2seq) prediction.

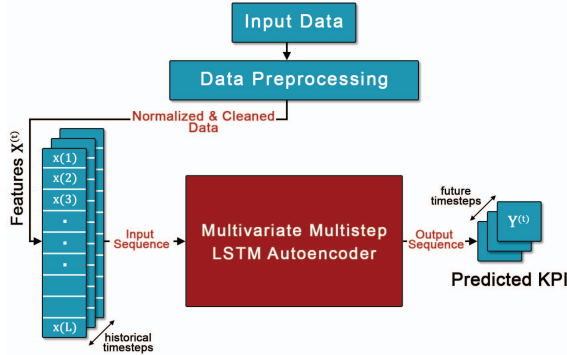


Fig. 1: QoS Prediction Scheme Overview

Figure 1 depicts an overview of the proposed scheme, which comprises a step-by-step methodology for the QoS prediction. The first step includes the aggregation and pre-processing of all the above-mentioned service and network-related features. This step is responsible for removing any potential irrelevant and/or redundant information present in the data set, as well as applying the $L2$ norm as normalization function. Each normalized feature vector is denoted by $X = \{x(1), x(2), \dots, x(L)\}$. Let $X(n)$, with $n \in [1, L]$, denote each normalized feature of the feature vector X , where L is the maximum number of features. The second step, provides a sequence of the normalized and filtered feature vectors, as input to the LSTM-based model. The length of the sequence is equal to a predefined historical time window (*historical timesteps*). The feature vector at timestep t is denoted by $X^{(t)}$. At the third step, the architecture of the QoS prediction model, employs 2 LSTM networks namely the encoder

and decoder. The encoder processes the input sequence and encoding it into a fixed-length vector (*context vector*). The decoder takes as input the encoded vector, reconstructs the sequence and outputs the predicted sequence of the target QoS metric, which is the ToD-UE's uplink throughput (ul_thr). As $Y = \{Y^{(t)}, Y^{(t+1)}, \dots, Y^{(t+H)}\}$, is denoted the predicted output sequence for multiple timesteps into the future (*future timesteps*), where H denotes the prediction time horizon.

IV. EVALUATION

A. Simulation Setup

This section presents the evaluation outcomes of the proposed scheme. The necessary data that were used for the evaluation were generated using the discrete-event network simulator (NS-3) [17]. The NS-3 LTE module is selected instead of the 5G-LENA [18] due to maturity in terms of implementation. However, the proposed prediction approach is not dependent on the radio technology and can be efficiently applied in 5G and beyond network environments. The simulated topology chosen for this study is an Urban Macro deployment for connected cars, known as "Manhattan Grid" and was considered from a use case defined in the 3rd Generation Partnership Project [19]. The 3GPP evaluation methodology defined in [20] was also used for the simulated environment. The simulated topology is depicted in Figure 2. The data produced by NS-3, simulate 3356 real-life seconds (~ 56 minutes) and are divided in multiple simulation runs with different configurations, concerning the different traffic load in the network. The sampling frequency of the data is set to 200 milliseconds (ms). There are 2 types of UEs in the simulation namely a single ToD-UE and multiple NonToD-UEs. The number of the NonToD-UEs corresponds the different traffic load levels that are introduced in the network. More precisely, there are 10 different traffic load levels, produced by 0, 5, 15, 30, 40, 50, 60, 70, 80 and 100 NonToD-UEs, respectively. Table I summarizes the NS-3 parameters used in the simulated scenario. The ToD-UE periodically generates traffic of 20 Mbps Uplink and 500 Kbps Downlink data rate, while the NonToD-UEs periodically generate traffic of 4 Mbps and 8 Kbps for the Uplink and Downlink data rate respectively. The ToD-UE is constantly moving in a circular route around the first eNB (eNB 1) with an average speed equal to 9.16 m/s (32.97 km/h) while the NonToD-UEs follow randomly generated trajectories inside the grid.

Parameter	Description	Default Value
# of eNBs		3
UEs' Transmission Power		23 dBm
eNB Transmission Power		40 dBm
eNB Center Frequency		2160 MHz
eNB Downlink and Uplink Bandwidth		20 MHz
eNB and UE antenna model type		1x Isotropic
Propagation Loss Model for LOS/NLOS		ITU-R P.1411

TABLE I: Parameters Used in the ns-3 Simulated Scenario

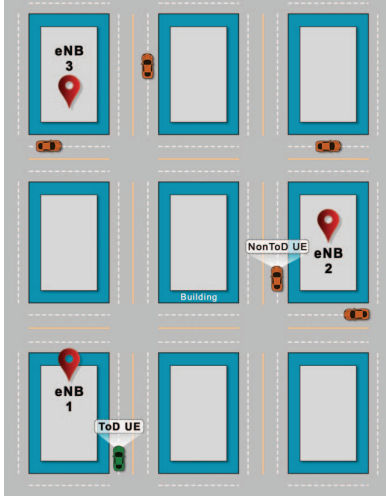


Fig. 2: NS-3 Virtual Topology

B. Experimental Results

In this last part, the results from the LSTM-based QoS Prediction scheme evaluation are presented. The batch size of the LSTM-based model was set to 64 and the total number of epochs to 500. In order to prevent any degradation in the performance of the validation set and as a result overfitting, an early stopping function was applied. This function monitors the Mean Squared Error (MSE) of the validation set and stops the training process when a minimum value is reached. To increase the performance of the model, the Adam optimizer was selected, with learning rate $lr = 0.0001$. L2 (weight decay) regularizer is applied to each hidden layer of 10^{-5} . Finally, the prediction time window was set to 7 seconds using a 21-second historical time window of monitored features as an input. The prediction process is performed every 1 second. More specifically, every 1 second the LSTM-based model predicts the ul_thr for the next 7 seconds. The split ratio for the dataset was set to 0.8, meaning that 80% of the available data was used for model training and the rest 20% for evaluation. As a result, 2658 seconds were used for the training phase and 698 for the evaluation.

Figure 3 illustrates the comparison between the groundtruth and the predicted output for the 7th second of each prediction time window. As it can be inferred from the figure the LSTM model is able to successfully capture the vast majority of the uplink throughput fluctuations of the ToD-UE.

The performance of the proposed LSTM-based model was evaluated with the use of two error metrics, namely the Normalized MSE (NMSE) and the Root MSE (RMSE) in *Mbps*. The NMSE and RMSE is calculated for each timestep for all prediction time windows. Since, the logging frequency of the data is set to 200 ms, each 7 second prediction time window consists of 35 timesteps ($\{t+200, t+400, \dots, t+7000\}$ in ms). The proposed model performs best from the 2nd second onwards with a slight increase at the last 2 seconds.

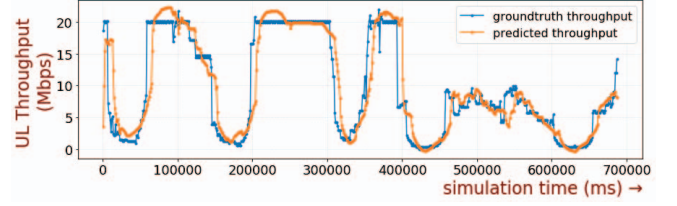


Fig. 3: LSTM Performance for the prediction horizon of 7 seconds

Error Metric	Overall Score	7th Second Score
NMSE	0.0539	0.0795
RMSE (Mbps)	2.458	3.458

TABLE II: Performance evaluation of the proposed LSTM-based scheme

As a result, Figure 4 illustrates the overall NMSE and RMSE (for the best scored timesteps - from the 2nd second onwards). The aforementioned values are summarized in Table II.

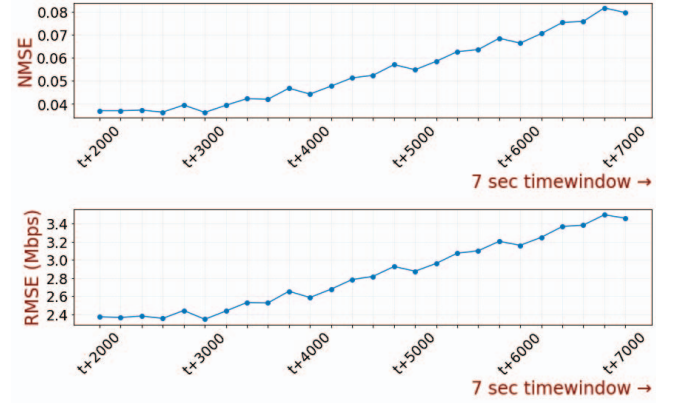


Fig. 4: NMSE (**top**) and RMSE (in Mbps) (**bottom**) for each timestep, from the 2nd second onwards, for all prediction time windows

Figure 5 depicts the distribution of the overall absolute error (AE), as well as the AE of the 7th second of all predicted time windows. In addition, for both abovementioned distributions, the 75th percentile is calculated, indicating that the 75% of the AE values will be less than 1.832 and 1.874 Mbps respectively.

V. CONCLUSION AND FUTURE WORK

In this paper, we investigated the the applicability and validity of an LSTM-based QoS prediction scheme, capable of processing ToD application- and network-oriented data towards identifying useful patterns in the form of time series, and predicting the uplink throughput QoS metric for future time horizons. The architecture of the LSTM-based mechanism was described, while a detailed evaluation section was discussed that indicated promising results for the specific use case. The future steps for this work will comprise firstly to

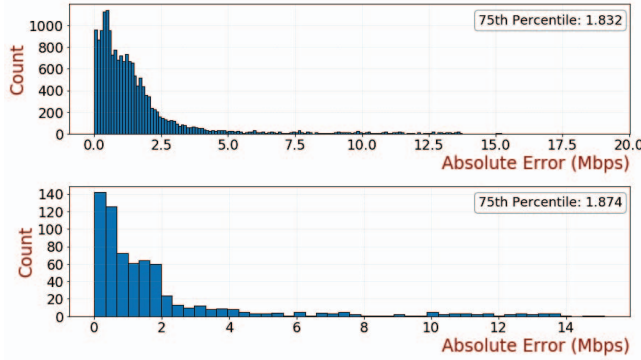


Fig. 5: Distribution and 75th percentile of the overall (top) and the 7th second (bottom) absolute error in Mbps

further investigate and enhance the architecture of the proposed Machine Learning framework, by integrating potential additional steps and modules that seem to further enhance the prediction outcomes; secondly, we plan to optimize the specific architecture, tailored for CAM-specific application data in order to identify whether the prediction accuracy can be further enhanced; thirdly, we plan to extend the predicted QoS metrics vector by targeting additional KPIs, such as end-to-end delay and packet loss; last but not least, we plan to experiment with larger-scale scenarios and for higher time horizons in the future, in order to come up with even more robust solutions.

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